Night-time moist heat driven by humidity, not temperature alone, is crossing physiologically anchored survival thresholds — with mortality risk attaching to the crossing event itself

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Abstract

Humid-heat risk assessments rest on daytime, intensity-based indices anchored to a theoretical 35 °C wet-bulb survival limit that laboratory physiology now shows to be too high. Using sub-hourly observations from 176 globally distributed stations (1976–2023), we show that night-time hours above a physiologically anchored 21.9 °C wet-bulb threshold have increased by +42.8 hours per decade, that warming alone explains none of this rise (humidity requirement +106%), and that climate models overstate the observed rise by 2.3–2.9× after optimal fingerprint scaling. In 225 European regions (77,551 region-weeks), mortality risk attaches to *crossing* the threshold rather than to its duration, so scaling risk by intensity growth (+47%) instead of extent growth (+26%) overstates burden 1.78×. Two wet-tropics cities provide independent sub-annual replication in which humidity, not dry-bulb heat, carries the signal. Absolute humid-heat thresholds saturate in the tropics, invalidating percentile-based warning triggers in 75.6% of European regions.

The theoretical wet-bulb temperature (T_w) limit of human survivability, 35 °C, has framed humid-heat risk assessment for over a decade[1]. Empirical physiology, however, places uncompensable heat stress far below this limit: updated survivability limits correspond to $T_w \approx 25.8$–34.1 °C for young adults and ≈ 21.9–33.7 °C for older adults[2], values that independent laboratory work continues to validate[3] and that are already being approached during present-day heat events[4]. Observational studies report that extreme humid heat has more than doubled in frequency since 1979[5], and detection and attribution analyses find anthropogenic fingerprints in heat-stress indices[6]. Yet the epidemiological record needed to translate these physical trends into health burdens is concentrated in

temperate, high-income settings, leaving the regions with the highest T_w severely under-represented[7] — a gap we address directly.

Two structural biases follow from this mismatch. First, health-relevant exposure has been measured with daytime, intensity-based indices: daily maximum T_w counts[5], gridded T_w extreme indices[8;9], or wet-bulb globe temperature (WBGT) products[10]. Whether night-time hours — when physiological recovery from daytime heat is expected[11;12] — are crossing low, physiologically anchored thresholds has not been quantified globally. Second, exposure metrics differ from the quantity that health models can validate: the optimal heat-stress metric for mortality modelling varies by country[13], station and reanalysis exposure data yield materially different risk estimates[14], and time aggregation alone biases heat-mortality burden estimates by 21.6% at the weekly scale[15]. The joint consequence — that the metric used for detection and projection is not the metric whose health relevance can be established — has not been examined for humid heat.

Here we close this loop with a single, physiologically anchored exposure metric applied consistently from global detection to European health attribution to wet-tropics replication. We define NWH_T as the annual number of night-time hours (solar elevation $< -0.833°$) with $T_w \geq T$, computed from sub-hourly station data (NOAA GHCNh) with T anchored to the 21.9 °C older-adult survivability lower bound[2] for the global analysis and to regionally applicable levels for the wet tropics. We quantify the trend and its attribution, test whether mortality risk follows the crossing event or its duration, decompose exposure extent versus intensity, and translate the results into observation-constrained projections, a graded warning scale, and diagnostics for current World Meteorological Organization/World Health Organization (WMO/WHO) warning guidance[16;17].

Results

Night-time humid-heat burden is rising on every usable threshold

Night-time hours above 21.9 °C averaged 1,546 h yr^{-1} over 1976–2023 across the 64 stations with ≥90% window coverage and rose by +42.84 h per decade (Fig. 1a; station median +18.27, sign-test $p = 1.0 \times 10^{-9}$, $n = 67$; population-weighted +52.13). The trend is positive in all eight analysis windows tested (+27.69 to +58.62 h per decade; Extended Data Fig. 6). The 21.9 °C threshold is the lowest at which the metric remains informative: 17.7% of station-years are zero and the mean has grown 1.22× from its 1976–85 level, whereas thresholds above 30 °C are floor-inflated (97.6% of station-years zero at 30.6 °C; 99.9% at 35 °C) and cannot measure trends (Fig. 1b). This zero-inflation structure is itself a result: the commonly used survivability framing measures exposure almost exclusively through its rare, daytime tail[18], whereas the night-time burden grows on thresholds an order of magnitude closer to physiological limits.

Humidity, not warming alone, drives the night-time rise

Counterfactual reconstructions that preserve variability and remove trends separately in temperature and humidity show that warming alone explains −6% of the $NWH_{21.9}$ rise (network-level S-calibre), whereas humidity change is required for +106% of it; the station-median decomposition gives +52% warming, +105% humidity and a zero interaction (Fig. 2a). An independent decomposition on a different station sample ($n = 67$) attributes 4% to temperature, 39% to humidity and 57% to their interaction. Under every alternative detrending calibre the humidity requirement remains 41–118% (Table 1). Subtropical stations carry the signal (humidity +48.9 h per decade), while equatorial stations show almost no trend because they are already saturated. Consistently, dry-bulb threshold-crossing hours rose by only +8.8 h per decade over the same period — 63% slower than the wet-bulb metric.

An anthropogenic fingerprint that models overstate

Optimal-fingerprint regression of observed $NWH_{21.9}$ trends (1976–2014) against CMIP6 historical and natural-only simulations yields a single-signal ALL scaling factor of $b_{\mathrm{ALL}} = +0.429 \pm 0.148$ (90% CI $[0.186, 0.671]$), with the natural-only signal rejected ($b_{\mathrm{NAT}} = -2.52 \pm 1.11$); the two-signal fit gives $b_{\mathrm{ALL}} = +0.348 \pm 0.168$ (Fig. 2b). Observed trends exceed internal variability by 3.08σ in the tropics and 1.36σ in the subtropics. Because the ALL response must be scaled down to match observations, unadjusted CMIP6 models overstate the observed rise by 2.3× (single-signal) to 2.9× (two-signal) (Fig. 2c) — consistent with known humid-heat biases in CMIP6[19]. All projections below are therefore scaled by the fingerprint factor before impact translation.

Mortality attaches to the crossing event, not to its duration

In 225 European NUTS2 regions with weekly mortality (77,551 warm-season region-weeks, 2001–2024), weeks in which nocturnal T_{w} crossed 21.9 °C in any hour had a death rate 1.0181× (95% CI $[1.0104, 1.0258]$, $z = +4.62$) that of non-crossing weeks, with an attributable fraction of 1.77% $[1.02, 2.52]$ (Fig. 3a). Controlling jointly for dry-bulb tropical nights *strengthens* the humid-heat association to 1.0292 $[1.0212, 1.0373]$, $z = +7.21$ — the humidity signal is not a dry-bulb alias. Binning weekly crossing hours reveals a step-shaped dose–response: risk is elevated for any crossing (0–2 h: 1.0156) and does not increase with additional hours (> 20 h: 0.9950, $z = -0.37$) (Fig. 3c). The association is present where the exposure has variance (southern/humid stratum 1.0036 per SD, $z = +4.75$) and absent where it does not (northern/cool 1.0009, contrast $z = +2.26$, $p = 0.024$; Fig. 3d). Aggregating the same panel annually attenuates the association by ∼93% (z from +34.4 to +1.1), extending the aggregation-bias evidence of Ballester

et al.[15] from the daily-to-weekly to the weekly-to-annual transition.

Extent, not intensity, governs risk — and intensity-based scaling overstates it

Between the first and last five years of 2000–2024, the European population exposed to nocturnal 21.9 °C crossings grew by +26% and the number of exposed regions by +29%, while mean crossing hours grew faster (+47%; $p = 0.0014$) (Fig. 5a). Because the dose–response is flat beyond the first crossing, risk follows extent, not intensity: scaling burden by the intensity trend overstates the risk-relevant growth by a factor of 1.78. Under observation-constrained SSP2-4.5 and SSP5-8.5 warming, the newly exposed European population reaches +78 and +146 million by end-century (unscaled: +166 and +237 million) (Fig. 5b), and each +1 °C of warm-season night-time T_{w} adds 42.3–70.0 million people newly exposed (region fixed-effect to pooled logistic dose–response; Fig. 5c). The model is calibrated to within −2.7% of the observed mean.

Wet-tropics replication, threshold failure and the power diagnosis

Two wet-tropics cities with sub-annual mortality — Hong Kong (132 warm-season months) and Singapore (264 months) — provide the independent replication that the literature lacks[7]; prior Hong Kong analyses relied on dry-bulb temperature indicators[20], whereas the signal documented here is carried by humidity. Monthly NWH_{25} is associated with mortality in each city (Hong Kong 1.0140 per SD, $z = +2.31$; Singapore 1.0073, $z = +1.69$) and in a random-effects meta-analysis of the dry-bulb-orthogonalised exposures (1.0030 $[1.0009, 1.0051]$, $z = +2.83$, $p = 0.0046$, $I^2 = 0\%$; Fig. 4a). A pre-specified negative control confirms the mechanism: the orthogonalised dry-bulb hot-night series carries *no* positive signal (Hong Kong $z = -2.57$, inverse; Singapore $z = +1.09$) — in these cities humidity, not dry-bulb heat, carries the mortality signal.

The same cities demonstrate why globally uniform thresholds fail in the tropics. In Singapore, the median monthly exceedance fraction is 1.000 at 21.9 °C and 0.981 at 24 °C: the exposure degenerates into night length, and an $\mathrm{NWH}_{21.9}$ mortality model returns a spurious $z = +8.61$ that we reject as a saturation artefact (Fig. 4b,c). Re-thresholding restores validity ($\mathrm{NWH}_{25.0}$: 1.0122, $z = +2.03$). A power diagnostic explains the long-standing absence of wet-tropics mortality evidence: Singapore's warm-season night-time T_{w} varies by only 0.50 °C (SD) against Hong Kong's 1.62 °C, so detecting the same per-degree effect requires 2.33× larger true effect sizes (minimum detectable effects 1.05 vs 2.45% per °C; Fig. 4d). The humid-heat evidence gap in the tropics is thus a quantifiable statistical-power problem, not evidence of absence.

Operational translation and a failure mode of current guidance

A three-level graded scale built on nocturnal T_w levels of 21.9, 24 and 25.8 °C is strictly monotonic in mortality across 225 European regions (level I 1.0591 $[1.0384, 1.0803]$, $z = +5.68$; level III 1.2686 $[1.1183, 1.4391]$, $z = +3.70$; trend 1.0577 per level, $z = +5.30$; Extended Data Fig. 8), and in the wet tropics must be replaced by local quantiles because absolute levels saturate. As a warning layer, the nocturnal trigger alone detects fewer event weeks than the current daytime rule (POD 0.102 vs 0.184) and is therefore positioned as an *add-on*: it flags 3,142 weeks that the daytime rule misses, with an event rate 1.53× the panel average. More consequentially, the WMO/WHO default of percentile-based triggers fails structurally under zero inflation: the 95th percentile of warm-season $NWH_{21.9}$ equals zero in 75.6% of European regions (63.6% at P97.5, 52.4% at P99) (Extended Data Fig. 9), so percentile rules cannot fire where they are most needed. Reference re-implementation at the two cities bounds the combined implementation uncertainty of the metric at $u_c = 35.7$–39.0%, dominated (22–32%) by the night definition, with wet-bulb formula choice contributing $< 0.6\%$ (Extended Data Fig. 7); the widely used one-third-rule approximation biases NWH_{25} by +12.1% and must not be used.

Discussion

Three findings reorganise how humid-heat health risk should be measured. First, nighttime threshold crossings — not daytime intensity — are where the physiologically relevant humid-heat signal has grown, and humidity change, not warming alone, is its driver. The metric family used in prior global assessments[5;8;21;22] counts rare daytime extremes; the same stations show the burden shifting into the night on thresholds within the experimentally validated survivability range[2;3]. Second, mortality risk in Europe attaches to the crossing event itself. This occurrence-type response conflicts with the adaptive-threshold behaviour reported for East Asian nighttime-only heatwaves, where risk rises only at high cumulative intensity[23]; the distinction is metric-based — cumulative excess heat versus physiological threshold crossing — and population- and scale-based, and it is directly falsifiable in cohorts with individual-level exposure. Third, because risk follows extent rather than intensity, impact assessments that scale burden by intensity trends overstate risk-relevant growth by 1.78× here; the correction is arithmetic, portable, and large.

Our results sharpen rather than soften the caveats. The wet-tropics replication rests on two high-income, high-adaptation cities; the power diagnosis quantifies why broader tropical evidence is absent but does not substitute for it. The full-sample exposure–response gradient across Europe is positive but is driven by regions whose exposure has no variance; restricted to informative regions the per-unit gradient is negative, consistent with adaptation along the humid-heat gradient (Extended Data Fig. 10). Heterogeneity

across European regions is substantial ($I^2 = 78\%$), the fixed-effect per-degree exposure response (42.3 million per +1 °C) should be treated as the conservative bound, and implementation uncertainty of 36–39% is dominated by the night definition — arguing for a standardised astronomical definition in future monitoring (Extended Data Fig. 11). The annual-aggregation diagnostic extends known aggregation biases[15] and reinforces that sub-annual mortality data are a prerequisite for humid-heat health assessment. Finally, the structural failure of percentile triggers under zero inflation is a direct, actionable input for revision of WMO/WHO warning guidance[16] and WBGT-based standards[10;17]: thresholds in hot-humid regions must be locally anchored, and the graded scale validated here provides that anchoring while remaining monotonic in mortality.

The observation-constrained projections — 42–70 million Europeans newly exposed per +1 °C of night-time humid heat, and +78 to +146 million by end-century under SSP2-4.5 and SSP5-8.5 — translate a physiologically anchored metric into policy-relevant numbers with the model bias removed rather than inherited[21;24]. Given that exposure metrics, not emissions scenarios, dominate the spread in heat-stress impact projections[19], adopting crossing-based, night-limited, locally anchored metrics is the single most consequential methodological change available to humid-heat impact assessment.

Methods

Data

Night-time exposure is computed from NOAA GHCNh sub-hourly station records (version 1.1.0): 176 stations with ≥20 qualifying station-years yield 4,164 of 4,763 downloaded station-years (87%) after quality control; 67 stations enter headline panels ($n = 64$ for the balanced 1976–2023 network-mean series). Station populations use GeoNames city points within 50 km with exponential decay to 300 km (9.01 billion person-equivalents of coverage). European weekly deaths are Eurostat NUTS3 weekly counts aggregated to 2021 NUTS2 (225 regions); gridded exposure uses NASA POWER (MERRA-2) with an Open-Meteo/ERA5 cross-check. City panels use Hong Kong Census and Statistics monthly deaths (Table 115-01022A) and Singapore SingStat registered deaths (M810121) with NASA POWER hourly meteorology. CMIP6 historical, SSP2-4.5, SSP5-8.5, DAMIP natural-only and piControl runs (tasmin, hurs) from five models (CNRM-CM6-1, CanESM5, HadGEM3-GC31-LL, IPSL-CM6A-LR, MRI-ESM2-0) follow Eyring et al.[25]. External validation uses HadEX3 TN90p and GISTEMP v4. WHO Mortality Database annual panels were used only for negative diagnostics and are reported in the Supplementary Information. Humidity's role in heat–health outcomes remains debated in the epidemiological literature[26]; distributed-lag and nonlinear exposure–response conventions follow Gasparrini[27].

Wet-bulb temperature and the NWH metric

T_{w} is computed per hourly record from dry-bulb temperature and dew point by iterative solution of the psychrometric equation (Stull and Davies-Jones implementations cross-validated to 0.05 °C; dew-point inversion as reference). Night is defined by solar elevation $< -0.833°$ computed from station coordinates, not by clock hour, because station timestamps are UTC. NWH_T counts night-time hours with $T_{\mathrm{w}} \geq T$ per station-year. Thresholds: 21.9 °C (older-adult survivability lower bound[2]) as headline; 23, 24, 25.8, 30.6 and 35 °C as the ladder; 25.0 and 25.8 °C for wet-tropics city models.

Trends, uncertainty and homogeneity

Trends use Theil–Sen regression with Mann–Kendall tests and Benjamini–Hochberg false-discovery control; spatial dependence is handled via Bretherton effective sample sizes and spherical k-means cluster bootstrap (headline 95% CI for the station-median trend $[+3.4, +57.0]$ h per decade). Relative homogeneity uses up to five reference neighbours within 1,500 km with Pettitt changepoint screening; homogenisation changes the trend ratio by 1.00. Urban-heat-island contamination is bounded by the most-rural quartile remaining significant ($p = 6.8 \times 10^{-4}$). Independent confirmation: HadEX3 TN90p versus night-recovery failure rate $r = 0.402$ ($p = 0.0069$; 92% sign agreement among non-zero stations).

Humidity necessity decomposition

Counterfactual series A (observed), B (temperature detrended), C (humidity detrended) and D (both detrended) are built with variance-preserving LOWESS detrending (S-calibre). Main effects are referenced to D: temperature $= \mathrm{tr}(B) - \mathrm{tr}(D)$, humidity $= \mathrm{tr}(C) - \mathrm{tr}(D)$, interaction $= \mathrm{tr}(A) - \mathrm{tr}(B) - \mathrm{tr}(C) + \mathrm{tr}(D)$. Network-aggregated (equal-weight annual means) and station-median variants are reported; linear and climatology-locked calibres are shown as sensitivities.

Detection and attribution

Optimal-fingerprint total-least-squares regression of the two-dimensional (tropical, subtropical) observed trend vector on ALL and NAT ensemble-mean responses, with noise covariance from 67 non-overlapping 39-year piControl segments; 90% confidence intervals are profiled. Projections are scaled by b_{ALL} (single signal) before impact translation; the transfer function from monthly mean/SD to crossing hours is validated against observed annual NWH ($R^2 = 0.9992$, bias $+0.2\,\mathrm{h\,yr^{-1}}$).

European health panel

Poisson generalised linear models with region and year fixed effects, Fourier seasonality (weekly), and region-clustered robust standard errors; exposures are raw binary (any nocturnal hour $\geq$ 21.9 °C), z-standardised continuous hours, or joint dry-bulb controls (tropical night, TX $\geq$ 30 °C). Regional estimates are pooled by DerSimonian–Laird random effects with empirical-Bayes shrinkage; heterogeneity is $I^2 = 78.2\%$ and is reported with the region-level effect distribution (median 1.021; 74.3% of regions > 1). Lag-1 displacement is null (0.9979), excluding harvesting. Gradient analyses regress regional slopes on background nocturnal T_{w} with cluster bootstrap, restricted and unrestricted (Extended Data Fig. 10).

Wet-tropics city panels and power

Monthly Poisson models with year splines (df 8), bimonthly Fourier terms, raw and dry-bulb-orthogonalised exposures; meta-analysis via random effects. Minimum detectable effects convert 80%-power detectable rate ratios per SD into per-degree equivalents. Negative control: orthogonalised dry-bulb hot nights ($\geq$ 28 °C).

Exposure extent, intensity and projections

Extent (share of population/regions with any crossing) and intensity (mean crossing hours) are compared over first-versus-last five-year means (2000–2024, 135 regions, 499 million people). The dose–response mapping uses a pooled logistic model calibrated to the observed mean, with a region fixed-effect alternative; scenario warming (CMIP6 multi-model mean, fingerprint-scaled) is applied to the fitted response.

Grading, warning skill and implementation uncertainty

The graded scale assigns European region-weeks to levels I–III by nocturnal T_{w} thresholds (21.9/24/25.8 °C) and evaluates monotonicity in the same Poisson framework; wet-tropics versions use within-city quantiles (cluster-robust SEs are unreliable with two clusters and are reported with HC0 intervals). Warning skill (POD, FAR, Heidke) compares four triggers on the same panel. Implementation uncertainty combines wet-bulb formula, night definition, input and instrument terms by root-sum-square from a reference re-implementation at both cities.

Code

All analyses are Python 3.13 (statsmodels, numpy, scipy, xarray). The full pipeline (steps 1–54), frozen result artefacts with SHA-256 hashes, and one-command repro-

duction of every figure and number in this Article are publicly available at `https://github.com/fanqimcse/nocturnal-moist-heat-burden` (tag v1.0.0; archival request accepted by Software Heritage; a DOI will be minted on acceptance).

Data availability

NOAA Global Historical Climatology Network-hourly (GHCNh) v1.1.0 (NCEI, https://www.ncei.noaa.g Eurostat weekly death statistics (https://ec.europa.eu/eurostat); NASA POWER (https://power.larc.na Open-Meteo ERA5 archive (https://open-meteo.com); Hong Kong Census and Statistics Department Table 115-01022A (https://www.censtatd.gov.hk); Singapore Department of Statistics M810121 (https://www.singstat.gov.sg); CMIP6 via the Earth System Grid Federation (https://esgf-node.llnl.gov); HadEX3 (https://www.climdex.org); GISTEMP v4 (https://data.giss.nasa.gov/gistemp); GeoNames (https://www.geonames.org); WHO Mortality Database (https://www.who.int/data/gho/data/themes/mortality-and-global-health-estimates). All datasets are public; no new empirical data were collected.

Code availability

The analysis pipeline, frozen artefacts (SHA-256 indexed), figure-generation code (`make_figures.py`) and a reproduction script for every reported number are available at the repository above and accompany the submission as Supplementary Software.

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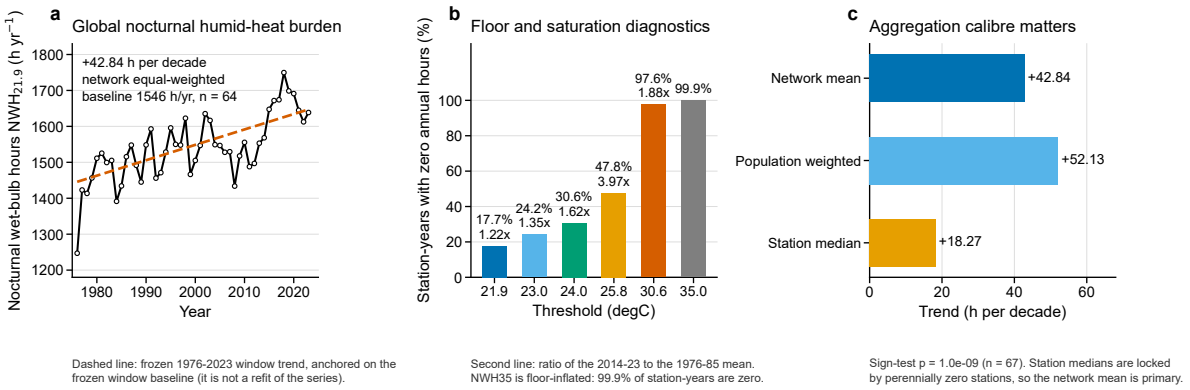
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Main figures and table



Dashed line: frozen 1976-2023 window trend, anchored on the frozen window baseline (it is not a refit of the series).

Second line: ratio of the 2014-23 to the 1976-85 mean. NWH35 is floor-inflated: 99.9% of station-years are zero.

Sign-test p = 1.0e-09 (n = 67). Station medians are locked by perennially zero stations, so the network mean is primary.

Figure 1: Global night-time humid-heat burden and the choice of threshold. **a**, Annual network-mean night-time hours with wet-bulb temperature $\geq$ 21.9 °C ($NWH_{21.9}$), 1976–2023, 64 stations with $\geq$90% window coverage; dashed line is the frozen window trend (+42.84 h per decade) anchored on the 1976–85 baseline (1,546 h yr^{-1}). **b**, Zero-inflation diagnostic: share of station-years with zero crossing hours (bars) and growth ratio of 2014–23 to 1976–85 means (labels) by threshold; thresholds above 30 °C are floor-inflated. **c**, Aggregation calibre matters: network mean (+42.84), population-weighted (+52.13) and station median (+18.27) trends on the identical station set; the station median is locked by permanently zero stations, so the network mean is primary.

Table 1: Primary results and key sensitivities. All values trace to frozen, hash-verified artefacts (Extended Data Table 2 gives the complete register, including retracted analyses that are never presented as results).

Quantity	Sample	Effect (95% CI unless stated)	Status
$NWH_{21.9}$ trend, 1976–2023	64 stations, network equal-weighted	+42.84 h per decade; baseline 1,546 h yr^{-1}	Primary
Window sensitivity (8 windows)	balanced sets per window	+27.69 to +58.62 h per decade, all positive	Sensitivity
Aggregation calibre	same 64 stations	+42.84 / +52.13 / +18.27 (mean / pop-weighted / median)	Sensitivity
Moisture necessity, $NWH_{21.9}$	72 stations, S-calibre	warming-only −6%; humidity requirement +106%	Primary
Optimal-fingerprint scaling	tropics + subtropics, 1976–2014	$b_{ALL} = +0.429 \pm 0.148$ (single); $+0.348 \pm 0.168$ (two-signal)	Primary
Mortality, crossing 21.9 °C (any219)	136 regions, 77,551 region-weeks	RR = 1.0181 [1.0104, 1.0258], $z = +4.62$, $I^2 = 78.2\%$; AF = 1.77% [1.02, 2.52]	Primary
+ dry-bulb tropical-night control	136 regions, joint model	any219 RR = 1.0292 [1.0212, 1.0373], $z = +7.21$	Identificati
Dose–response bins	64 regions with graded exposure	0–2 h 1.0156; >20 h 0.9950 ($z = -0.37$)	Step respo
South–north Europe contrast	101 / 35 regions	1.0036 vs 1.0009 per SD; contrast $z = +2.26$, $p = 0.024$	Effect mod
Wet-tropics meta (orthogonalised)	HK 132 + SG 264 months	1.0030 [1.0009, 1.0051], $z = +2.83$, $I^2 = 0\%$	Replicatio
Threshold saturation, Singapore	300 months	exceedance fraction 1.000 (21.9 °C), 0.672 (25.0 °C)	Threshold
Minimum detectable effect, tropics	HK / SG	1.05 vs 2.45% per °C (ratio 2.33×)	Power diag
Extent vs intensity, Europe	135 regions, 2000–2024	population +26%, regions +29%, NWH +47%; overstatement 1.78×	Primary
Newly exposed population	135 regions, 499 million	SSP2-4.5 +78 M; SSP5-8.5 +146 M (end-century, scaled)	Primary
Exposure response per +1 °C	same sample	70.0 M pooled / 42.3 M fixed effects	Primary
Implementation uncertainty	reference re-implementation	$u_c = 35.7$–39.0%; night definition dominates	Limitation

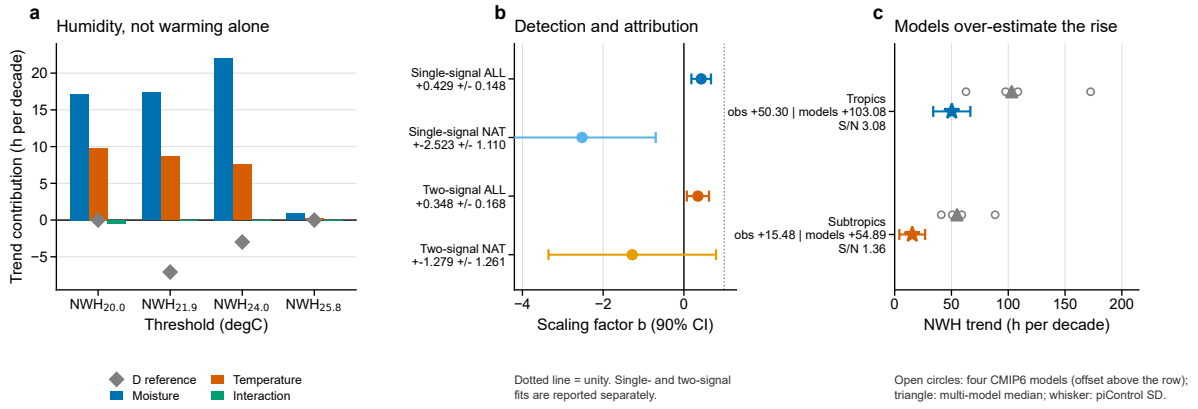


Figure 2: Humidity necessity and anthropogenic detection. **a**, Variance-preserving counterfactual decomposition of the $\mathrm{NWH}_{21.9}$ trend by threshold (station-median calibre, $n = 72$): moisture (blue), temperature (orange), interaction (green), double-detrended reference (diamond). **b**, Optimal-fingerprint scaling factors with 90% confidence intervals: single-signal ALL ($+0.429 \pm 0.148$), single-signal NAT (rejected), two-signal ALL ($+0.348 \pm 0.168$) and two-signal NAT. **c**, Observed versus CMIP6 NWH trends (h per decade): open circles, individual models; triangles, multi-model median; whisker, piControl internal variability; observation-to-internal-variability ratios 3.08 (tropics) and 1.36 (subtropics).

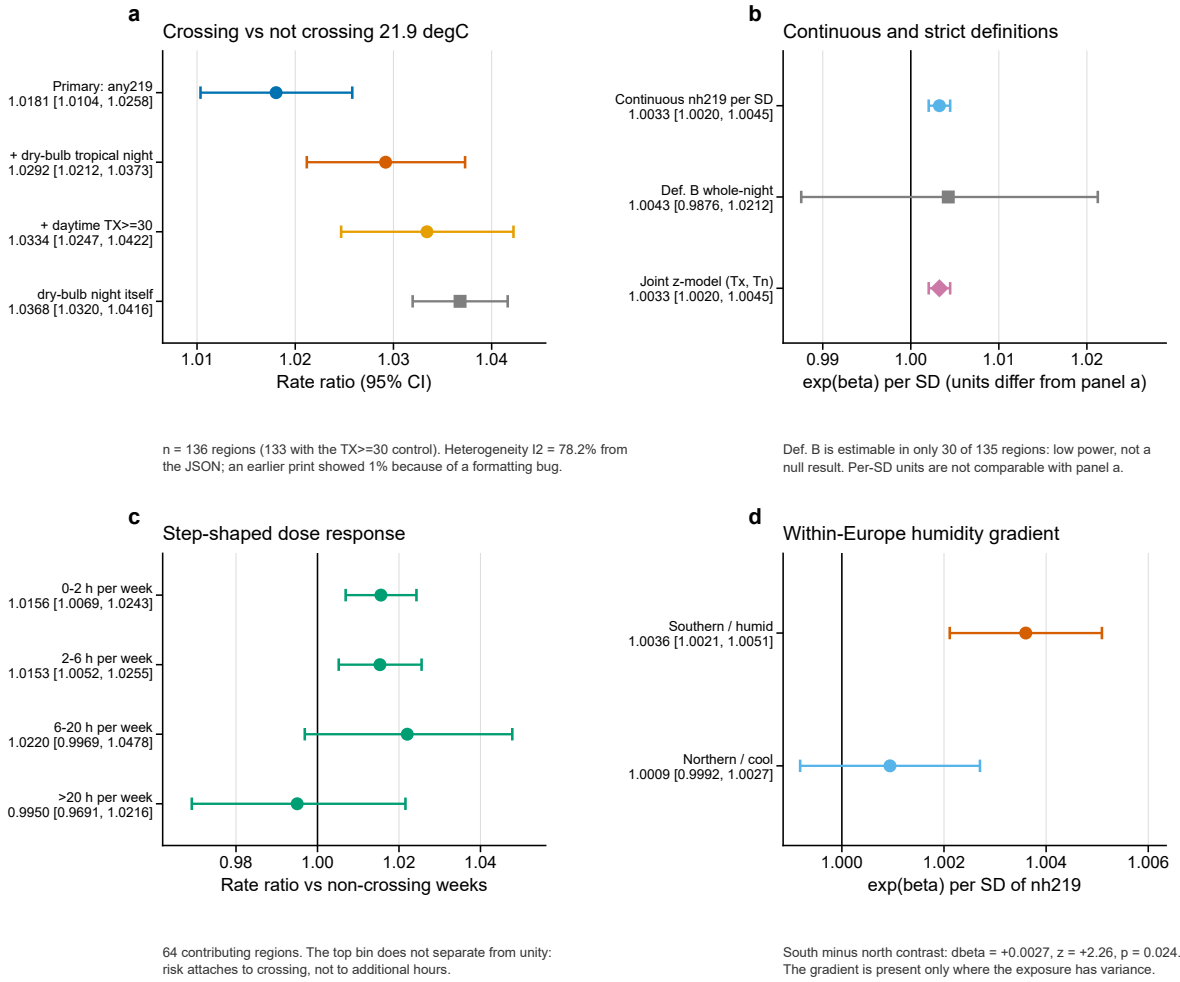


Figure 3: Mortality attaches to the crossing event (Europe, weekly, 2001–2024). **a**, Rate ratios for weeks with any nocturnal hour ≥ 21.9 °C, alone and adjusted for dry-bulb tropical nights or daytime TX ≥ 30 °C (136 regions; 133 with the TX control). **b**, Continuous (per SD) and strict whole-night definitions; definition B is estimable in only 30 regions (low power, not a null result). **c**, Dose–response by weekly crossing hours: risk attaches to the first crossing and does not increase with additional hours. **d**, Southern/humid versus northern/cool strata (per SD of continuous exposure); contrast $z = +2.26$, $p = 0.024$.

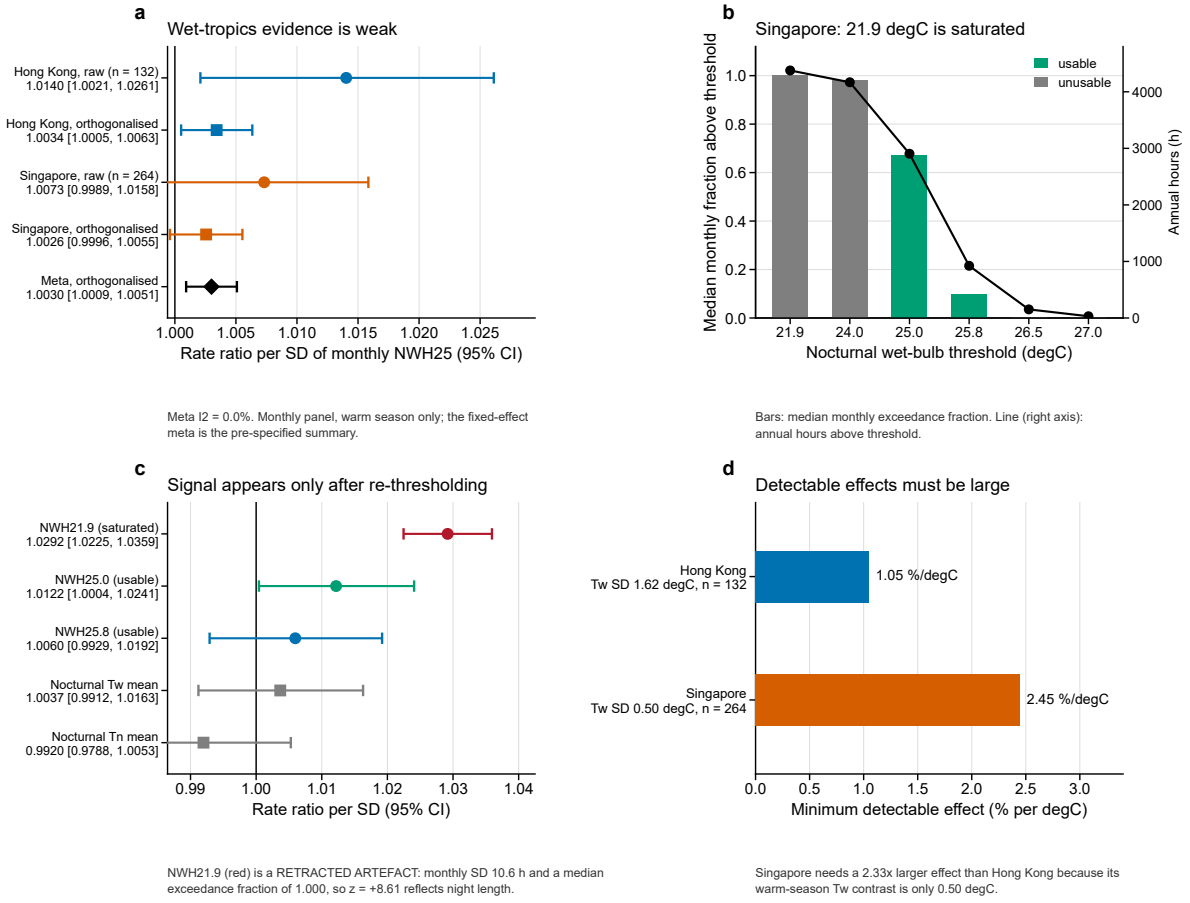


Figure 4: Wet-tropics replication, threshold failure and statistical power. **a**, Monthly NWH_{25} mortality associations (rate ratio per SD, 95% CI) for Hong Kong ($n = 132$ months) and Singapore ($n = 264$), raw and dry-bulb-orthogonalised, with the pre-specified random-effects meta-analysis ($I^2 = 0\%$). **b**, Singapore threshold saturation: median monthly exceedance fraction (bars; usable in green) and annual crossing hours (line, right axis); 21.9 and 24 °C are saturated. **c**, Re-thresholded models: the $\mathrm{NWH}_{21.9}$ result (red) is a rejected saturation artefact (median exceedance fraction 1.000; $z = +8.61$ reflects night length); 25.0 and 25.8 °C are valid. **d**, Minimum detectable effects: Singapore requires a 2.33× larger true effect than Hong Kong because its warm-season night-time T_{w} varies by only 0.50 °C (SD) versus 1.62 °C.

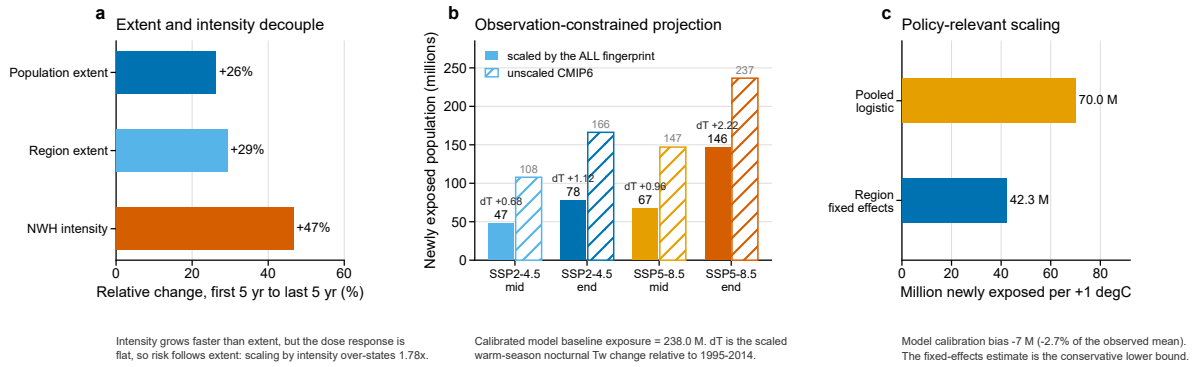
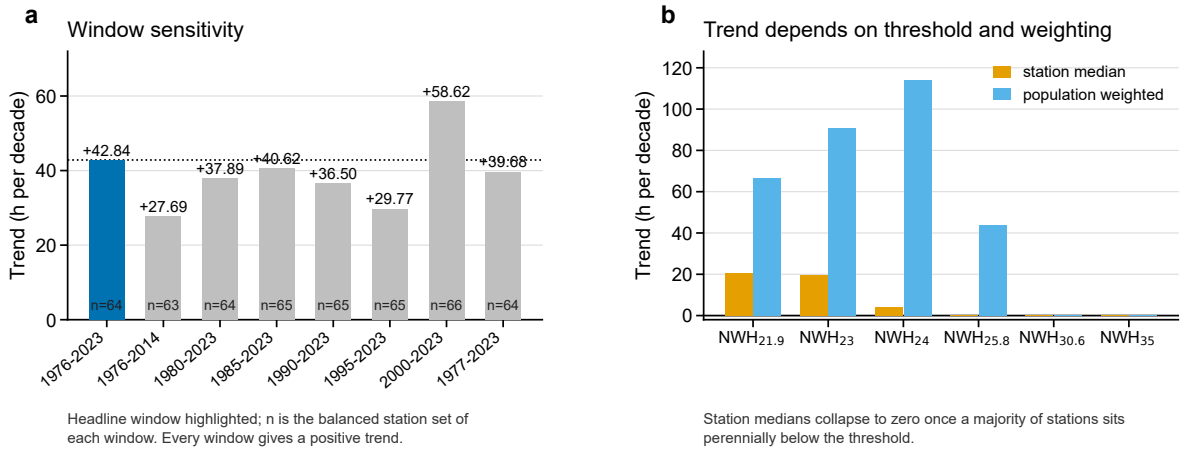


Figure 5: Extent–intensity decoupling and observation-constrained projections. a, Relative change between the first and last five years of 2000–2024 in exposed population (+26%), exposed regions (+29%) and mean crossing hours (+47%); because the dose–response is flat beyond the first crossing, scaling risk by intensity overstates risk-relevant growth 1.78×. **b**, Newly exposed European population (millions) under SSP2-4.5 and SSP5-8.5 at mid- and end-century, fingerprint-scaled (solid) versus unscaled CMIP6 (hatched); ΔT is the scaled warm-season night-time T_{w} change relative to 1995–2014. **c**, Policy-relevant scaling: millions newly exposed per +1 °C of warm-season night-time T_{w} (pooled logistic 70.0; region fixed effects 42.3, conservative); model calibration bias −7 million (−2.7% of the observed mean).

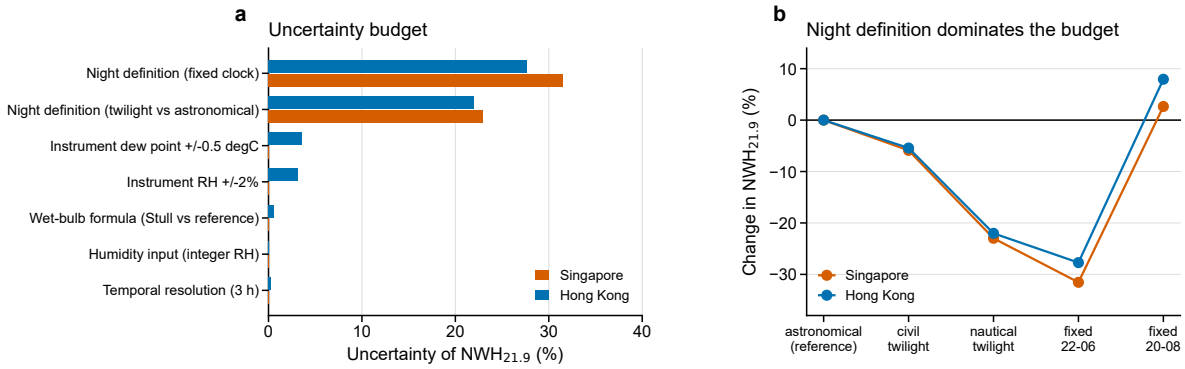
Extended Data



Headline window highlighted; n is the balanced station set of each window. Every window gives a positive trend.

Station medians collapse to zero once a majority of stations sits perennially below the threshold.

Figure 6: Window robustness of the headline trend. Theil–Sen trend of $NWH_{21.9}$ for eight start/end windows (1976–2023 headline); all windows are positive, range +27.69 to +58.62 h per decade.



Combined root-sum-square u_c: Singapore 39.03%, Hong Kong 35.72%.

Each point is the percentage change in NWH21.9 relative to the astronomical-night reference implementation.

Figure 7: Implementation uncertainty budget. Component-wise and combined (root-sum-square) relative uncertainty of the metric from a reference re-implementation at Hong Kong ($u_c = 35.72\%$) and Singapore ($u_c = 39.03\%$); the night definition dominates (22–32%), wet-bulb formula choice contributes $< 0.6\%$.

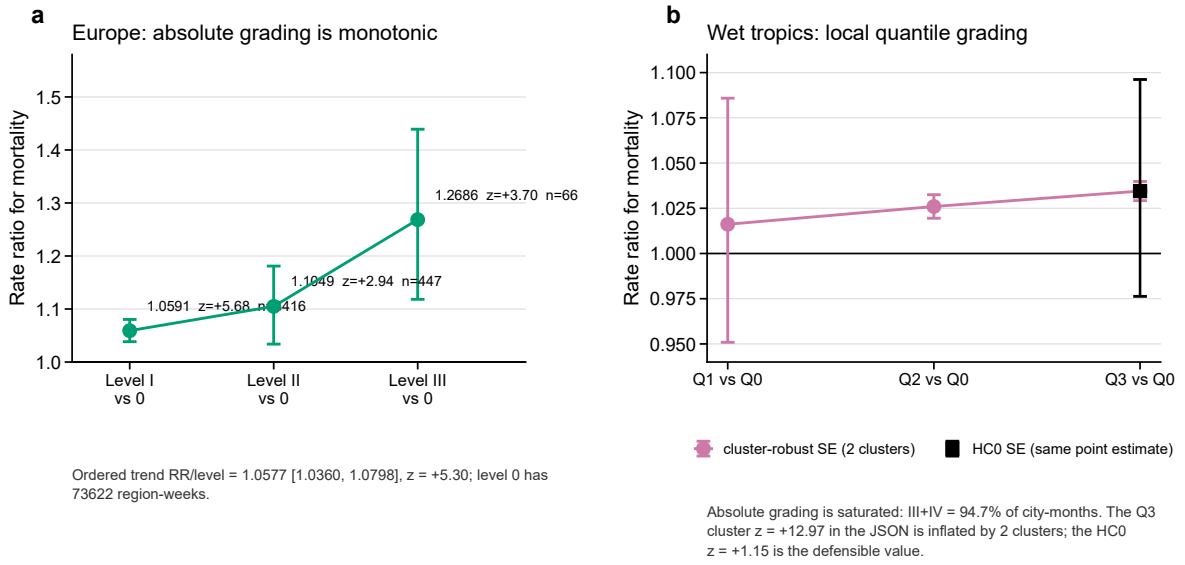


Figure 8: Graded scale validation. **a**, Mortality by graded level in Europe (225 regions, 77,551 region-weeks): I 1.0591 [1.0384, 1.0803], II 1.1049 [1.0338, 1.1810], III 1.2686 [1.1183, 1.4391]; monotone trend 1.0577 per level, $z = +5.30$. **b**, Local quantile grading in the wet tropics: with only two cities, cluster-robust SEs are unreliable and HC0 intervals are reported alongside (Q3 HC0 CI [0.9763, 1.0962]).

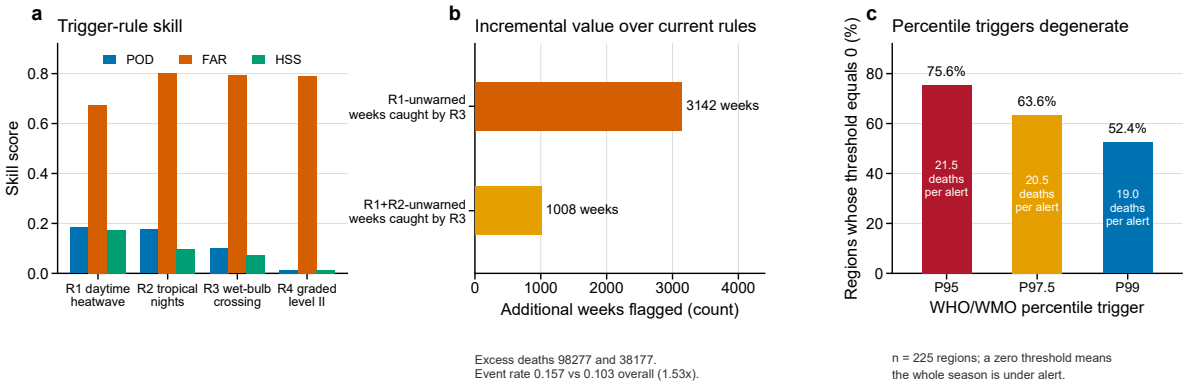


Figure 9: Warning-skill evaluation and the percentile-trigger failure. **a**, POD, FAR and Heidke skill for four triggers; the nocturnal trigger alone (POD 0.102) is below the current daytime rule (0.184) and is positioned as an add-on. **b**, Incremental value: 3,142 weeks flagged only by the nocturnal trigger, event rate 1.53× the panel average. **c**, Zero-inflation failure of the WMO/WHO percentile default: the P95 (P97.5, P99) warm-season $\mathrm{NWH}_{21.9}$ threshold equals zero in 75.6% (63.6%, 52.4%) of European regions.

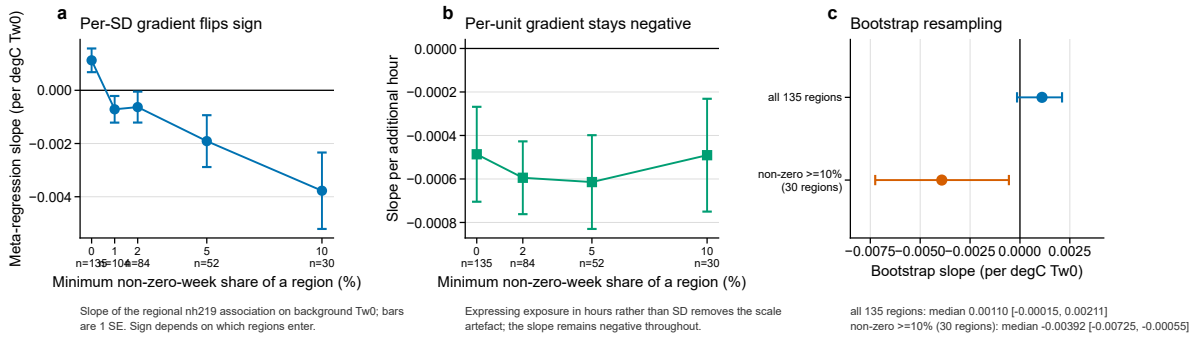


Figure 10: Exposure–response gradient robustness. Meta-regression of regional slopes on background nocturnal T_w: the unrestricted all-region slope is positive (+0.00113 per SD) but is driven by zero-variance regions; restricted to regions with non-zero exposure in $\geq$ 10% of weeks the slope is negative (−0.00377); per-unit slopes are negative throughout. Bootstrap intervals shown.

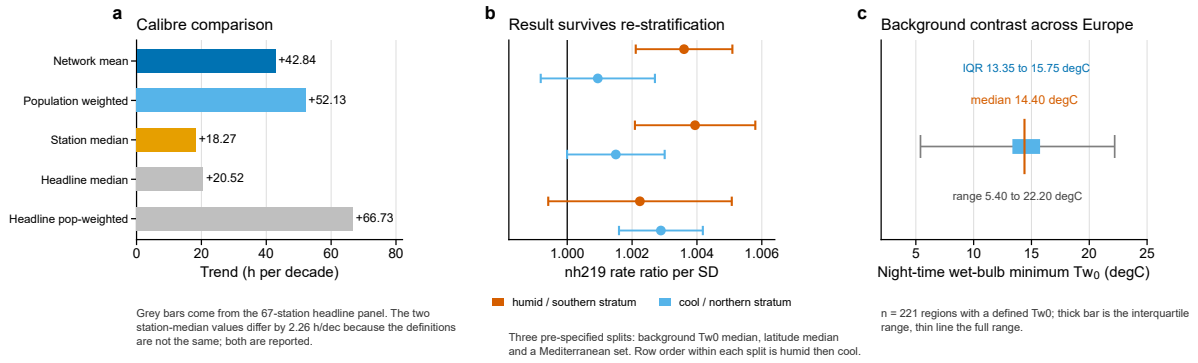


Figure 11: Calibre and stratification robustness. a, Two station-median constructions of the headline trend (20.52 vs 18.27 h per decade) are plotted side by side as different estimators. **b**, The south–north contrast is stable across three stratification schemes (background T_w, latitude, Mediterranean): humid/southern strata 1.0036–1.0022 per SD versus cool/northern 1.0009–1.0015. **c**, Distribution of background warm-season hottest-night wet-bulb minima across 221 regions (median 14.40 °C, IQR 13.35–15.75 °C).

Table 2: Complete register of frozen results, samples, specifications and status. Every numeric cell traces to a hash-verified artefact (see figures_manifest.csv). Rows R1–R6 are retracted, excluded or corrected items that are never presented as positive results.

#	Fig.	Quantity	Sample	Effect (95% CI unless stated)	Status
1	Fig1a	NWH21.9 trend, 1976–2023	64 stations, ≥90% coverage, network equal-weighted	+42.84 h per decade; baseline 1,546 h yr^{-1}	Primary
2	Fig1b	Zero-inflation by threshold	176 stations screened, 67 retained	NWH21.9: 17.7% zero station-years, ratio 1.22×; NWH35: 99.9% zero	Limitation
3	Fig1c	Aggregation sensitivity	identical 1976–2023 station set	+42.84 / +52.13 / +18.27 h per decade	Sensitivity
4	Fig2a	Moisture vs warming decomposition	72 stations, counterfactuals A/B/C/D	moisture +17.31 h/dec (+105%); warming +8.60 (+52%); interaction 0.00	Primary
5	Fig2b	Fingerprint ALL scaling factor	tropics + subtropics, 1976–2014; piControl $n = 67$ segments	single +0.429 ± 0.148; two-signal +0.348 (90% CI +0.072 to +0.624)	Primary
6	Fig2c	Observed vs modelled trends	67 stations; 4 CMIP6 models	tropics obs +50.30 vs models +103.08 (S/N 3.08); subtropics +15.48 vs +54.89 (1.36)	Supporting
7	Fig3a	Mortality, crossing 21.9 °C (any219)	136 regions, 77,551 region-weeks	RR = 1.0181 [1.0104, 1.0258], z = +4.62, I^2 = 78.2%; AF = 1.77% [1.02, 2.52]	Primary
8	Fig3a	+ dry-bulb tropical-night control	136 regions, joint binary model	any219 RR = 1.0292 [1.0212, 1.0373], z = +7.21; dry-bulb RR = 1.0368 [1.0320, 1.0416]	Identification
9	Fig3b	Continuous specification	135 regions, non-zero variance	RR per SD = 1.0033 [1.0020, 1.0045], z = +5.24	Sensitivity
10	Fig3b	Strict whole-night definition	30 regions with such events	RR = 1.0043 [0.9876, 1.0212], z = +0.50	Low power
11	Fig3c	Dose–response bins	64 regions, graded weekly exposure	0–2 h 1.0156 [1.0069, 1.0243]; 2–6 h 1.0153; 6–20 h 1.0220; >20 h 0.9950 [0.9691, 1.0216]	Step response
12	Fig3d	South vs north Europe	south 101 / north 35 regions	south 1.0036 [1.0021, 1.0051]; north 1.0009 [0.9992, 1.0027]; contrast z = +2.26 ($p = 0.024$)	Supporting
13	Fig4a	Wet-tropics association (NWH25)	HK 132; SG 264 months	HK 1.0140 (z = +2.31); SG 1.0073 (z = +1.69); meta (orthogonalised) 1.0030 [1.0009, 1.0051], $I^2 = 0\%$	Replication
14	Fig4b	Threshold saturation, Singapore	300 months, 2001–2025	21.9 °C fraction 1.000; 24.0 0.981; 25.0 0.672; 25.8 0.098	Limitation
15	Fig4c	Re-thresholded models, Singapore	monthly panel, n = 300	NWH25.0 1.0122 (z = +2.03); NWH25.8 1.0060 (z = +0.89); NWH21.9 1.0292 (z = +8.61) REJECTED	Artefact + result
16	Fig4d	Minimum detectable effect	HK 132; SG 264 months	HK 1.05% per °C; SG 2.46% per °C (ratio 2.33×)	Power limitation
17	Fig5a	Extent vs intensity decoupling	135 regions, 2000–2024	population +26%, regions +29%, NWH +47%; intensity-based scaling overstates 1.78×	Primary
18	Fig5b	Newly exposed population	135 regions, 499 million	SSP2-4.5 end-century +78 M (unscaled 166); SSP5-8.5 +146 M (unscaled 237)	Primary
19	Fig5c	Exposure response per +1 °C	same sample, calibrated	70.0 M (pooled) / 42.3 M (fixed effects)	Primary
20	ED2a	Implementation uncertainty budget	reference re-implementation, two cities	u_c = 39.03% (SG), 35.72% (HK); night definition dominates	Sensitivity
21	ED3a	Graded scale, Europe	225 regions, 77,551 region-weeks	I 1.0591 [1.0384, 1.0803]; II 1.1049; III 1.2686; trend 1.0577/level	Monotonicity
22	ED3b	Local quantile grading, tropics	two cities, 300 city-months	Q2 1.0260; Q3 1.0345 with 2-cluster caveat; HC0 CI for Q3 [0.9763, 1.0962]	Caveated
23	ED4a	Warning trigger skill (R1–R4)	77,551 region-weeks	R3 POD = 0.102 below R1 POD = 0.184; positioned as add-on	Supporting
24	ED4b	Incremental nocturnal value	same panel	3,142 weeks (98,277 deaths); event rate 0.157 vs 0.103 (1.53×)	Supporting
25	ED4c	WHO/WMO percentile-trigger failure	225 regions; P95/P97.5/P99	P95 = 0 in 75.6% of regions (63.6%, 52.4%); capture 21.5/20.5/19.0 deaths per alert	Guidance limit
26	ED5	Gradient robustness	135 estimable regions of 225	slope +0.00113 (all) vs −0.00377 (non-zero ≥ 10%); per-unit slope negative	Sensitivity
27	ED6b	Three stratification schemes	225 regions, 77,551 region-weeks	humid/southern 1.0036/1.0039/1.0022; cool/northern	Supporting